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TERM END EXAMINATIONS (TEE) – October 2026

Programme	:	**B. Tech.**	**Semester**	:	**Fall Semester 2026-27**
Course Name	:	**Database Management Systems**	**Course Code**	:	**CSE3001**
Date/Session	:	**15/10/2026 / Session-II**	**Slot**	:	**A11+A12+A13**
Time	:	**3 Hrs.**	**Max. Marks**	:	**100**

Answer ALL the Questions

Q. No.		Question Description	Marks
		PART A – (60 Marks)	
1	(a)	A university has several departments. Each department offers multiple courses and has several faculty members. A student can enroll in multiple courses, and each course can have many students. Each student is assigned to one department. Identify the entities, attributes, relationships, cardinalities, and participation constraints, and draw a complete ER diagram for the university database.	12
		OR	
	(b)	Explain the three-level architecture of a Database Management System (DBMS) with a neat diagram. Discuss the concept of data abstraction, mappings, and data independence.	12
2	(a)	Given the relations EMPLOYEE (EID, Name, Dept, Salary) and DEPARTMENT (Dept, Dept_Name), formulate relational algebra expressions to find employees earning more than ₹50,000, employees working in the CSE department, department names along with employee names, and employees whose salary is greater than the average salary of their department.	12
		OR	
	(b)	Consider the relation R(Student, Course, Instructor) with the functional dependencies (Student, Course) → Instructor and Instructor → Course. Explain the concept and conditions of Boyce-Codd Normal Form (BCNF). Determine the candidate key(s) of the given relation and check whether the relation satisfies BCNF. Identify the functional dependency that violates BCNF, if any, and decompose the relation into appropriate BCNF relations. Justify the decomposition and verify that the resulting relations satisfy BCNF.	12
3	(a)	Using the STUDENT (Student_ID, Student_Name, Dept, Marks) and DEPARTMENT (Dept_ID, Dept_Name) tables, formulate SQL queries to retrieve students scoring above 75 marks, display students department-wise in descending order of marks, calculate the average marks for each department, and identify departments having an average mark above 70. Use suitable SELECT, WHERE, ORDER BY, GROUP BY, and HAVING clauses.	12
		OR	

	(b)	Differentiate between Data Definition Language (DDL), Data Manipulation Language (DML), and Transaction Control Language (TCL). Write the syntax and suitable examples of CREATE, ALTER, DROP, INSERT, UPDATE, DELETE, COMMIT, ROLLBACK, and SAVEPOINT commands.	12
4	(a)	Develop a PL/SQL program using an explicit cursor to retrieve employee records department-wise and calculate the total and average salary for each department. Include suitable cursor operations and exception handling.	12
		OR	
	(b)	A disk-based database contains a relation with a large number of records. Compare ordered indexing, B+ tree indexing, static hashing, and dynamic hashing in terms of search cost, insertion, deletion, and storage requirements. Recommend a suitable indexing technique for a frequently updated database and justify your choice.	12
5	(a)	Explain the ACID properties of transactions with suitable examples. Discuss atomicity, consistency, isolation, and durability, and describe how a DBMS ensures atomicity and durability in the presence of transaction failures.	12
		OR	
	(b)	Clarify the Two-Phase Locking (2PL) protocol and its variants. Discuss how 2PL guarantees conflict serializability and examine the problems of deadlock and cascading rollback associated with lock-based concurrency control.	12
		PART B – (40 Marks)	
6		What is a weak entity set? Explain its characteristics and illustrate it with a suitable ER diagram.	8
7		**Enlighten Codd's rules for relational database systems. Also discuss entity integrity** and referential integrity constraints with suitable examples	8
8		Describe SQL triggers and their applications in database systems. Discuss BEFORE and AFTER triggers with a suitable example of a trigger used to maintain data integrity or an audit record.	8
9		Distinguish between PL/SQL procedures and functions. Write the basic structure of each and illustrate their use with suitable examples.	8
10		Define the Two-Phase Commit (2PC) protocol. Explain the roles of the coordinator and participants during the prepare and commit phases, and discuss the failure situations that may arise during the protocol.	8

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